

CE 310 — Week 3 Assignment: Conditional Probability and Engineering Decisions

Due: Wednesday of Week 4 by 9:00 AM — submit to D2L

Overview

This assignment extends the Week 3 lab into new territory. You will compute conditional probabilities using new variables, thresholds, and combined conditions that were not covered in lab, interpret those results through an engineering lens, and explain in writing why conditioning on context changes what a probability means for design.

How to Submit

☐ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

Requirement	Detail
File format	.xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF
Sheet name	Must be exactly Answers — do not rename or delete it
Major declaration	Enter CE or ArcE in cell B4 — this determines which 2b row is graded
Column layout	Enter your formula (or, for Problem 3, your arithmetic) in Column C and the resulting value in Column D — do not insert, delete, or move columns
Formulas	Type them as text — put an apostrophe before the =, e.g. '=COUNTIF(B2:B8761,">90")', so Excel stores the formula instead of running it. Screenshots are not accepted
Written responses	Type them in the yellow cells lower down the same sheet — do not submit a separate PDF
File name	CE310_W3_<NetID>.xlsx (e.g. CE310_W3_abc123.xlsx)
Submission	Upload the .xlsx file to D2L Dropbox before the deadline

Problem 1 — New Conditional Probability Computations (30 pts)

The Week 3 lab focused on cooling demand and electricity conditioned on July, summer, weekdays, and business hours. This problem asks you to compute three probabilities not found in the lab — using new seasons, new thresholds, and a new energy column.

All three parts use the same dataset: Week3_Tucson_Office_Energy_2023.csv

Part 1a — Cooling Demand in Spring (10 pts)

Calculate:

$P(\text{Cooling} > 30 \text{ kWh} \mid \text{Season} = \text{Spring})$

Using the definition of conditional probability:

$$P(A \mid B) = \frac{\text{Count}(A \text{ AND } B)}{\text{Count}(B)}$$

In Excel, the numerator and denominator are each a COUNTIFS.

Note: Use the Season column (column E), not the Month column. For comparison: $P(\text{Cooling} > 30 \mid \text{Summer}) = 56.66\%$ (you can verify with a COUNTIFS on “Summer”).

Show both COUNTIFS formulas and the resulting probability. In 1–2 sentences, compare your result to the Summer reference value above and explain physically why cooling demand above 30 kWh is so much more frequent in Summer than in Spring.

Enter your response in the P1a_written cell of your Answer Sheet.

Part 1b — Heating Electricity in Winter (10 pts)

The lab examined cooling demand (column H, Cooling_kWh). This part turns to a different column: HeatingElec_kWh (column I), which records electricity consumed by electric resistance heating.

Calculate:

$P(\text{HeatingElec} > 5 \text{ kWh} \mid \text{Season} = \text{Winter})$

Note: This uses column I (HeatingElec_kWh), not column H (Cooling_kWh). The threshold 5 kWh is not arbitrary — it corresponds roughly to a small supplemental resistance heater running at partial load, which is meaningful to distinguish from trace heating.

Show both COUNTIFS formulas and the resulting probability. In 2–3 sentences, discuss what this value means for a building engineer: during what fraction of winter hours is meaningful electric heating occurring, and why might the fraction be less than 100% even in a cold season in Tucson?

Enter your response in the P1b_written cell of your Answer Sheet.

Part 1c — Peak Electricity During Summer Business Hours (10 pts)

Your lab established that business hours and season are both strong predictors of electricity consumption. This part asks you to combine both conditions at a higher threshold — one that captures the most energy-intensive hours of operation.

Calculate:

$P(\text{Electricity} > 140 \text{ kWh} \mid \text{Season} = \text{Summer AND BusinessHours} = \text{Yes})$

This requires a three-column COUNTIFS.

Note: For reference, $P(\text{Electricity} > 140 \text{ kWh} \mid \text{BusinessHours} = \text{Yes}) = 59.19\%$ — the single-condition value before adding the Summer filter.

Show all formulas. In 2–3 sentences, interpret the result: how much does adding the Summer condition change the probability relative to the single-condition BusinessHours value above, and what does this suggest about the concentration of extreme electricity demand in summer business hours?

Enter your response in the P1c_written cell of your Answer Sheet.

Problem 2 — Cooling System Sizing and Peak Month Demand (18 pts)

Your lab results for the summer and peak-month cooling picture: - $P(\text{Cooling} > 50 \text{ kWh} \mid \text{Summer}) = 37.82\%$ (lab cell B11) - Month with highest $P(\text{Cooling} > 50) = \text{June}$ (lab cell B12) - Mean Cooling during Summer + BusinessHours = 74.48 kWh (lab cell B13)

This problem uses those values to reason about HVAC sizing, then adds one new computation — different for each major — at a higher threshold.

2a. ASHRAE 90.1-2022 (Energy Standard for Buildings Except Low-Rise Residential Buildings) requires that cooling systems be sized for peak load conditions — not average seasonal load. Using the lab values above, explain in 2–3 sentences why mean cooling

during summer business hours (74.48 kWh) is an insufficient basis for sizing a cooling plant. What does $P(\text{Cooling} > 50 \mid \text{Summer}) = 37.82\%$ add to that picture?

Enter your response in the P2a_written cell of your Answer Sheet.

2b — answer ONLY the version matching your major. Declare CE or ArcE in cell B4 of your Answer Sheet; that declaration determines which of the two rows below is graded. Leave the other blank.

2b (Civil Engineering). A demand-response contract prices summer weekday peaks separately, so the utility needs to know how often the building draws more than 150 kWh in an hour during summer weekdays — the threshold above which the peak tariff applies.

Calculate:

$P(\text{Electricity} > 150 \text{ kWh} \mid \text{Season} = \text{Summer AND DayType} = \text{Weekday})$

Note: This is a two-condition COUNTIFS on Season (column E) and DayType (column D), with the threshold on Electricity (column G).

CE students: enter this in the P2a row of your Answer Sheet.

2b (Architectural Engineering). June was identified as the peak month for high-cooling-demand probability. For HVAC sizing purposes, an engineer wants to know how often cooling demand exceeds a higher threshold of 75 kWh specifically in June — the level at which a standard commercial rooftop unit would be operating near capacity.

Calculate:

$P(\text{Cooling} > 75 \text{ kWh} \mid \text{Month} = \text{June})$

Note: Month is stored as a number (column B). Use 6 for June, not the text “June.”

ArcE students: enter this in the P2b row of your Answer Sheet.

Show all formulas and report the result.

2c. In 3–4 sentences, synthesize three values into a recommendation for sizing this building’s cooling system: your own 2b result, the lab’s mean cooling of 74.48 kWh (B13), and the lab’s $P(\text{Cooling} > 50 \mid \text{Summer}) = 37.82\%$ (B11). Would sizing the system for exactly the mean load (74.48 kWh) be adequate? Reference ASHRAE 90.1’s requirement that cooling systems be sized for peak load conditions, not annual average load. CE students: frame the recommendation around when the peak tariff threshold is crossed; ArcE students: frame it around June rooftop-unit capacity.

Enter your response in the P2c_written cell of your Answer Sheet.

Problem 3 — Bayes' Theorem: HVAC Fault Detection (20 pts)

A chilled-water plant serving a Tucson office building is monitored by an automated diagnostic system that flags anomalous condenser water delta-T values. Based on commissioning records and three years of maintenance logs, the following probabilities are known:

Known quantity	Value
$P(\text{Fouling})$ — fraction of summer operating hours with significant chiller tube fouling	0.12
$P(\text{Flag} \mid \text{Fouling})$ — probability the diagnostic flags the hour when fouling is present	0.88
$P(\text{Flag} \mid \text{No Fouling})$ — false positive rate	0.18

Part 3a — Compute $P(\text{Fouling} \mid \text{Flag})$ Using Bayes' Theorem (12 pts)

A flag has just fired. What is the probability that the chiller actually has significant fouling?

Step 1: Compute $P(\text{Flag})$ using the total probability rule:

$$P(\text{Flag}) = P(\text{Flag} \mid \text{Fouling}) \cdot P(\text{Fouling}) + P(\text{Flag} \mid \text{No Fouling}) \cdot P(\text{No Fouling})$$

Step 2: Apply Bayes' theorem:

$$P(\text{Fouling} \mid \text{Flag}) = \frac{P(\text{Flag} \mid \text{Fouling}) \cdot P(\text{Fouling})}{P(\text{Flag})}$$

In your work, label which value is the prior, which is the likelihood, and which is the posterior.

Show all arithmetic. Report your final answer as a percentage rounded to one decimal place.

Enter $P(\text{Flag})$ in the P3a_PFlag row and $P(\text{Fouling} \mid \text{Flag})$ in the P3a_posterior row of your Answer Sheet. Put the arithmetic itself in Column C of each row and the value in Column D.

Part 3b — Verify With a Frequency Table (5 pts)

Imagine 1,000 summer operating hours. Fill in the following table using the given probabilities:

	Flag fires	No flag	Row total
Fouling present	?	?	?
No fouling	?	?	?
Column total	?	?	1,000

Compute $P(\text{Fouling} \mid \text{Flag})$ from the table as: $(\text{Fouling AND Flag}) / (\text{all Flag})$. Confirm it matches your Part 3a result (within rounding).

Fill in the P3b_fouling, P3b_nofouling and P3b_total rows of your Answer Sheet. The grand total (1,000) is already entered for you.

Note: Fractional hours are correct here — the question says “imagine 1,000 hours”, so 105.6 is a valid cell value. Do not round to whole hours.

Part 3c — Engineering Interpretation (3 pts)

In 2–3 sentences: the diagnostic system detects 88% of fouled hours, yet the posterior you computed in Part 3a is far lower than that. Explain why these two numbers are consistent — cite your own Part 3a value — and state what maintenance protocol this suggests: should a flagged alarm trigger an emergency shutdown, a scheduled inspection, or something in between?

Enter your response in the P3c_written cell of your Answer Sheet.

Problem 4 — Reflection (12 pts)

Answer in one paragraph (100–150 words):

A colleague — a project manager with an engineering background — reviews your Week 3 lab summary and says: “Conditional probability is just a technicality — if 14% of all hours have high cooling demand, I’ll just size the system for 14% load frequency. The conditional breakdowns by season or business hours are interesting but they don’t change the bottom line.”

Using at least one specific number from your lab or this assignment, write a professional response to this claim. Your response should address whether the 14% unconditional figure captures what a building systems engineer actually needs to know for design, and what the conditional values reveal that the marginal probability obscures.

Enter your response in the P4_written cell of your Answer Sheet.

Submission Requirements

- Type all written responses in the designated yellow cells of your Assignment Answer Sheet. Submit CE310_W3_<NetID>.xlsx to D2L.
 - Type each COUNTIFS formula as text in Col C, next to the value it returned in Col D. Screenshots are not accepted.
 - Problem 3 must show all intermediate arithmetic in the Answer Sheet itself: Steps 1 and 2 in the P3a_PFlag and P3a_posterior rows, the full frequency table in the P3b_* rows, and typed prose in the P3c_written cell.
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Grading Summary

Problem	Points
Problem 1a — $P(\text{Cooling} > 30 \mid \text{Spring})$: formulas, result, 1–2 sentence comparison	10
Problem 1b — $P(\text{HeatingElec} > 5 \mid \text{Winter})$: formulas, result, 2–3 sentence interpretation	10
Problem 1c — $P(E > 140 \mid \text{Summer AND BusinessHours})$: 3-condition COUNTIFS, result, interpretation	10
Problem 2 — Cooling System Sizing and Peak Month Demand (2a written, 2b by major, 2c written)	18
Problem 3a — Bayes calculation: $P(\text{Flag})$, $P(\text{Fouling} \mid \text{Flag})$, all arithmetic shown	12
Problem 3b — Frequency table: correct cell values, correct posterior from table	5
Problem 3c — Interpretation: explains base rate effect, states appropriate maintenance response	3
Problem 4 — Reflection paragraph (100–150 words, at least one specific number cited)	12
Total	80 pts

This assignment is worth 80 raw points and will be scaled to the Post-Lab Assignment category in the course gradebook.